

KUNAL TOMAR

Indore, Madhya Pradesh

☎ (+91) 8770782647 ✉ kunaltomarmu26@gmail.com [in kunal-tomar1](https://www.linkedin.com/in/kunal-tomar1) [CodeKunalTomar](https://github.com/CodeKunalTomar) kunaltomar.dev

Summary

Final-year Computer Science student specializing in Artificial Intelligence, with hands-on experience in ROS2-based autonomous systems, scalable software development, and machine learning pipelines. Skilled in Python, C++, computer vision, algorithms, and deep learning frameworks (TensorFlow, PyTorch). Developed end-to-end projects in multi-robot coordination, UAV navigation, and real-time AI applications, demonstrating reliability, innovation, and technical leadership. Strong coding standards, analytical reasoning, and research-driven problem solving applied to deliver high-quality solutions in complex, interdisciplinary environments.

Education

Medicaps University

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.81/10

August 2022 – April 2026 (expected)

Indore, MP

Relevant Coursework

Java Programming (CS3CO32)

Advanced Java Programming (CS3CO37)

Mobile Virtual Reality and Artificial Intelligence (NPTEL – IIT Mandi (noc25-cs80))

Technical Skills

Languages: Python, C++, C, SQL, R

Frameworks & Platforms: ROS2, TensorFlow, PyTorch

Libraries: NumPy, SciPy, Pandas, OpenCV, Matplotlib, scikit-learn

Simulation & Robotics: PyBullet, Gazebo, PX4, RViz2, Unity 3D

Tools: Git, GitHub, Jupyter, MySQL, VirtualBox, Google Colab, RStudio, VS Code, Anaconda

Operating Systems: Linux (Ubuntu), Windows

Experience

Indian Institute Of Information Technology Allahabad

AI and Robotics Intern

June 2025 – August 2025

Prayagraj, UP

- Developed and validated an autonomous UAV navigation system for GPS-denied environments, implementing real-time obstacle detection and path planning; achieved <1 m navigation accuracy in validation tests.
- Integrated PX4 Autopilot with ROS2 using Offboard Mode to enable 10 Hz trajectory control and seamless communication between planning, perception, and control nodes.
- Engineered sensor-noise filtering and safety-override mechanisms using OpenCV and C++, reducing false positives and ensuring <100 ms emergency response.
- Utilized Python, C++, ROS2, PX4, Gazebo, TensorFlow, PyTorch, and Git to design, test, and optimize AI-driven robotic systems at IIIT Allahabad's Center for Intelligent Robotics (CIR).

Projects

Autonomous Multi-Robot Coordination System with Real-Time Computer Vision and ROS2 Integration

July 2025

Python, C++, ROS2, OpenCV, PyBullet

- Designed and developed an autonomous multi-robot coordination system integrating ROS2-based communication, real-time computer vision, and inverse kinematics for synchronized control of dual 7-DOF robotic arms in a PyBullet simulation environment.
- Implemented a multi-camera vision pipeline using OpenCV, Hough transforms, and Bayesian confidence scoring, enabling robust object detection, environment perception, and real-time spatial awareness.
- Developed a ROS2 communication architecture with custom message definitions and optimized Quality-of-Service (QoS) configurations to ensure reliable and synchronized multi-robot operation.
- Engineered a damped least squares inverse kinematics solver and collision-avoidance framework, achieving smooth trajectory planning, precise end-effector control, and safe cooperative motion between robots.

OptiConnect: Connect-4 AI with Non-Blocking Web UI (Web Workers)

December 2024

JavaScript, HTML5, CSS3 — [Live Demo](#)

- Built a browser-based 7×7 Connect-4 game with an AI opponent, using JavaScript and Web Workers to enable concurrent computation while keeping the interface smooth and responsive.
- Designed a modular architecture separating UI handling and background computation — the main thread managed user interactions and animations, while a worker thread handled game logic and minimax evaluation via postMessage/onmessage communication.

- Implemented a depth-limited minimax algorithm with randomized tie-breaking to balance decision quality and performance for complex board states.
- Optimized for large search spaces with efficient tree traversal, and structured the system to support future alpha-beta pruning and heuristic-based evaluation functions.
- Delivered a polished user experience through animated chip drops, hover feedback, real-time win highlighting, and robust input handling for invalid or full moves.

Leadership / Extracurricular

- Earned an **Elite + Gold Certificate (Top 1%)** in NPTEL's *Mobile Virtual Reality and Artificial Intelligence* course by IIT Mandi with a consolidated score of **93%** [\[Certificate\]](#).
- Led a 3-member team "Hidden Layers" to a top **4%** finish (Rank **3,270** of **82,790** teams) in the Amazon ML Challenge 2025.
- Achieved Rank **19** (top **0.5%**) in CodeKaze 2023 (Coding Ninjas – Round 2), a national-level competitive programming contest.